CLINICAL · NOSE & SINUSES

Skull base and the parapharyngeal space

The skull base is the floor of the brain and the roof of the nose, sinuses, nasopharynx, ear and neck. Tumours here present to the ENT surgeon as a blocked nose, a unilateral middle ear effusion, facial numbness, diplopia, hoarseness or a lump behind the tonsil. Modern skull base surgery is a team effort with neurosurgeons. Endoscopic endonasal routes now reach the sella, clivus and much of the anterior skull base without a facial incision. The parapharyngeal space, just below, has a logic of its own: decide whether the mass lies in front of or behind the styloid.

AT A GLANCE

- Abducens runs free in the cavernous sinus: often the first nerve to fail.
- Prestyloid mass is usually pleomorphic adenoma; poststyloid is neural or paraganglioma.
- Never biopsy a parapharyngeal mass through the mouth.
- Nasoseptal flap is pedicled on the posterior septal artery.
- Multilayer closure; vascularised nasoseptal flap for high-flow leaks; lumbar drains used selectively.

1 CHAPTER ONE Applied anatomy

Three fossae and the spaces below.

1.1 Anterior skull base

The anterior skull base separates the frontal lobes from the nose, ethmoid sinuses and orbits. In the midline lies the *cribriform plate*, its thinnest part, perforated by the olfactory fila; beside it the thin *lateral lamella* is a common site of CSF leak. The *crista galli*, where the falx attaches, rises in front, and the *planum sphenoidale* lies behind. Laterally are the orbital roofs and the *fovea ethmoidalis* of the frontal bone, with the anterior and posterior ethmoidal arteries crossing the ethmoid roof.

1.2 Middle cranial fossa and the central skull base

The middle fossa holds the temporal lobes. Centrally, the body of the sphenoid contains the sphenoid sinus, with the *sella turcica* above it holding the pituitary and a *cavernous sinus* on each side. In the lateral wall of the cavernous sinus run III, IV, V1 and V2, while VI runs free within the sinus beside the internal carotid artery, which is why VI is so often the first nerve to fail.

The foramina of the region carry predictable contents: the superior orbital fissure (III, IV, V1 and VI), the optic canal, the foramen rotundum (V2), the foramen ovale (V3), the foramen spinosum (the middle meningeal artery), and the foramen lacerum with the carotid canal. The **clivus** slopes from the dorsum sellae to the foramen magnum; its upper part is the basisphenoid and its lower part the basiocciput.

Seen from the sphenoid sinus, the carotid and optic nerve make bulges in the lateral wall, separated by the *optico-carotid recess*, and the bone over the carotid is dehiscent in some people. An **Onodi cell** is a posterior ethmoid cell that extends above the sphenoid and may contain the optic nerve.

1.3 Posterior fossa and lateral skull base

The petrous temporal bone contains the cochlea, labyrinth, facial nerve and internal carotid artery. The anterior surface of the petrous apex forms the floor of the middle fossa, with the arcuate eminence, the greater superficial petrosal nerve and *Kawase's triangle*, and its posterior surface carries the internal auditory canal. The jugular foramen transmits IX, X and XI with the jugular bulb, and the hypoglossal canal carries XII below and medial to it.

1.4 Infratemporal and pterygopalatine fossae

The **infratemporal fossa** lies below the middle cranial fossa, behind the maxilla and deep to the ramus, and contains the pterygoid muscles, V3, the maxillary artery and the pterygoid venous plexus.

The **pterygopalatine fossa** is a small inverted cone behind the maxillary sinus that opens in every direction: to the orbit through the inferior orbital fissure, to the

infratemporal fossa through the pterygomaxillary fissure, to the nose through the sphenopalatine foramen, to the middle cranial fossa through the foramen rotundum and vidian canal, and to the mouth through the greater palatine canal. It is therefore the crossroads by which juvenile angiofibroma and perineural cancer spread.

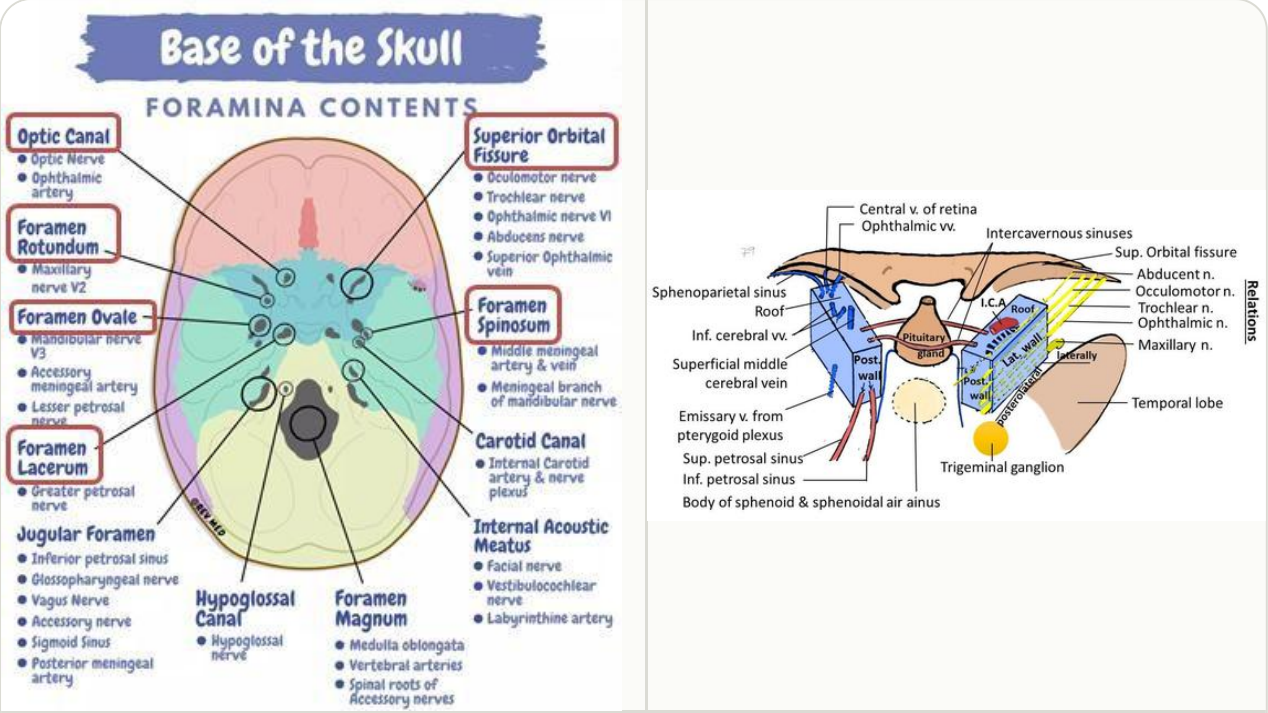


FIGURE 1 Foramina of the skull base and their contents, and the cavernous sinus with its contents.

Anatomy atlas · ENT Steps p. 152, Doss p. 85

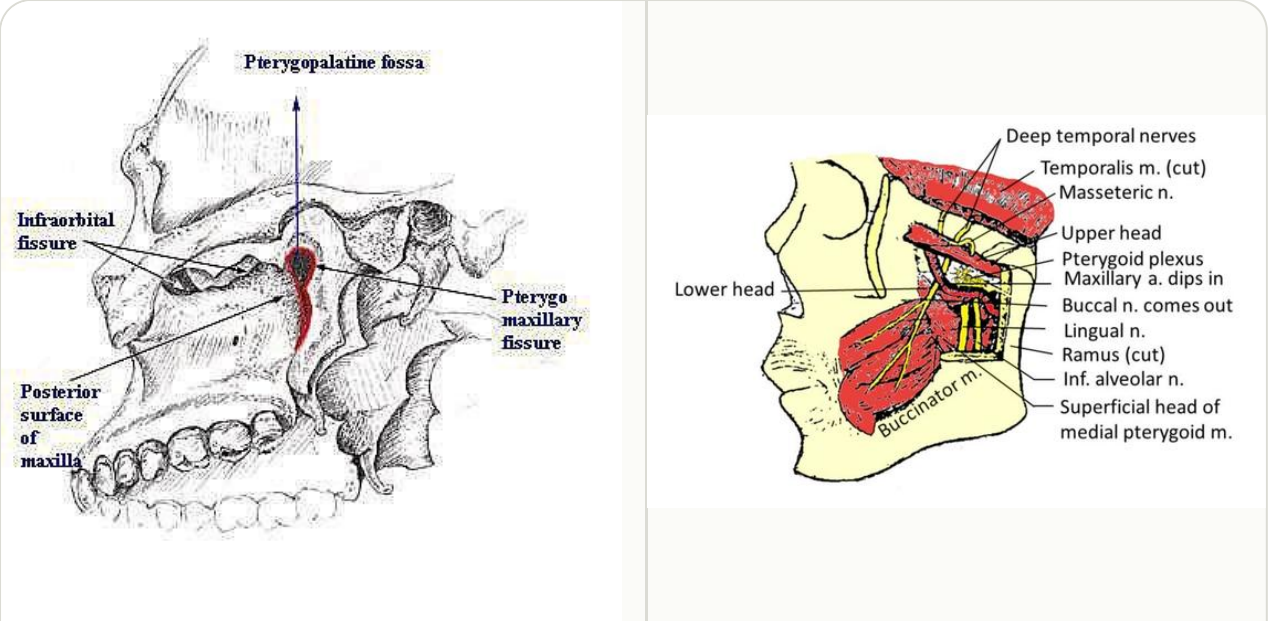


FIGURE 2 The pterygopalatine fossa, and the infratemporal fossa after removal of the ramus.
Anatomy atlas · ENT Steps p. 160, Doss p. 54

CHAPTER TWO

2 The parapharyngeal space

An inverted pyramid.

2.1 Boundaries and compartments

The parapharyngeal space is an inverted pyramid with its base on the skull base and its apex at the greater cornu of the hyoid. Medially lie the buccopharyngeal fascia and superior constrictor, and so the tonsil; laterally the medial pterygoid, the ramus and the deep lobe of the parotid; posteriorly the prevertebral fascia; and anteriorly the pterygomandibular raphe.

The *tensor–styloid fascia*, formed by the styloid process and its muscles, divides the space into two compartments. The **prestyloid compartment** contains fat, the deep lobe of the parotid, minor salivary rests, the maxillary artery and branches of V3. The **poststyloid (carotid) compartment** contains the internal carotid artery, the internal jugular vein, nerves IX to XII, the sympathetic chain and nodes.

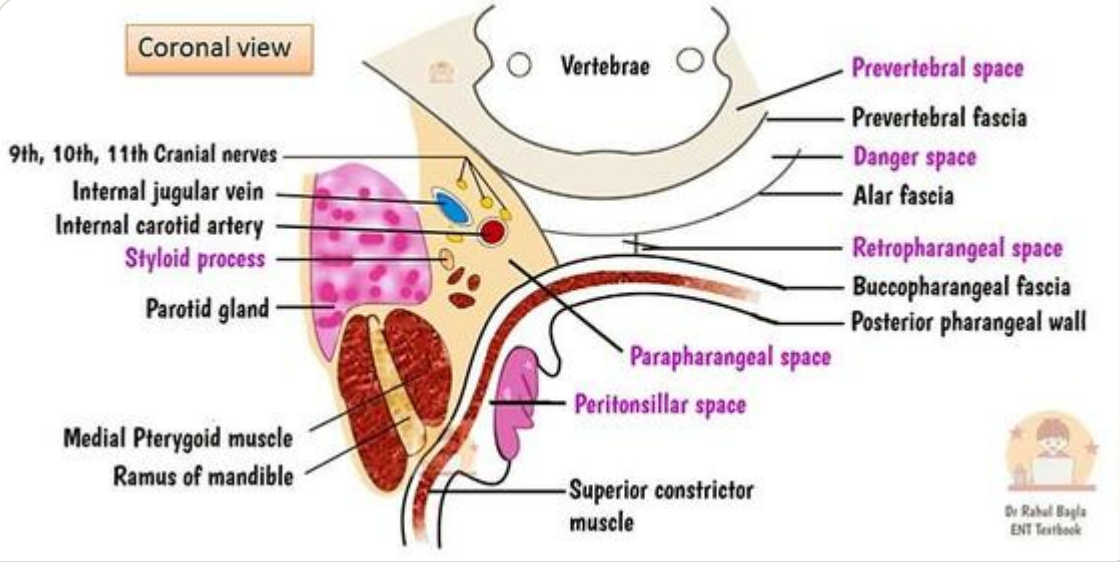


FIGURE 3 The parapharyngeal and retropharyngeal spaces.
Anatomy atlas · ENT Steps p. 198

2.2 Tumours

PARAPHARYNGEAL SPACE TUMOURS BY COMPARTMENT

COMPARTMENT	COMMON TUMOURS	IMAGING HINT
Prestyloid	*Pleomorphic adenoma* (deep lobe or minor salivary), other salivary tumours, lipoma	Fat plane between tumour and parotid lost (deep lobe origin) or preserved (minor salivary); carotid pushed *posteriorly*
Poststyloid	*Schwannoma* (vagus, sympathetic chain), *paraganglioma* (vagal, carotid body), neurofibroma, nodes (metastases, lymphoma), meningioma	Carotid pushed *anteriorly* or splayed; flow voids in paragangliomas

About half of parapharyngeal tumours are salivary, mostly pleomorphic adenoma, a fifth to a third are neurogenic, and the rest are paragangliomas and others; about 80% are benign. They are often silent until large, then appear as a smooth bulge pushing the *tonsil and soft palate medially* under normal mucosa, or as fullness behind the angle of the jaw, with a muffled voice, sleep apnoea, Eustachian tube obstruction or dysphagia. Cranial nerve palsies, Horner syndrome or pain suggest malignancy or a neural tumour.

MRI, with CT, places the tumour in its compartment and shows the vessels; metanephrines are measured whenever a paraganglioma is possible, and prestyloid

and nodal masses can be sampled by image-guided FNA. A *transoral biopsy is never done*, because it seeds a pleomorphic adenoma and makes a paraganglioma bleed.

The **transcervical** approach through the submandibular triangle, dividing the stylomandibular ligament, is the workhorse. Deep lobe tumours need a **transparotid–transcervical** approach, large tumours reaching the skull base a **transcervical approach with mandibulotomy**, and selected prestyloid tumours can be removed by **transoral robotic** surgery. The old blind transoral enucleation risks rupture, bleeding and incomplete excision.

OPERATIVE GUIDES IN THEATRE 3 operations

Tumours of the prestyloid parapharyngeal space, the transoral approach, and excision of neural tumours.

Management of Tumors in the Prestyloid Parapharyngeal Space
Prestyloid parapharyngeal
Transoral Approach to the Parapharyngeal Space
Excision of Neural Tumors of the Parapharyngeal Space
Neural tumours

3 CHAPTER THREE Skull base tumours

What arises here.

3.1 Anterior skull base

The malignant tumours of the anterior skull base are mostly sinonasal: squamous carcinoma, adenocarcinoma in woodworkers, adenoid cystic carcinoma, *olfactory neuroblastoma*, *sinonasal undifferentiated carcinoma*, neuroendocrine carcinoma and mucosal melanoma, together with sarcomas and lymphoma. The benign lesions are meningioma, inverted papilloma, juvenile angiofibroma, fibro-osseous lesions, mucoceles and encephaloceles.

3.2 Sella and parasellar region

The **pituitary adenoma** is the commonest sellar tumour; microadenomas are under 10 mm and macroadenomas larger. Mass effect produces *bitemporal hemianopia*,

headache and hypopituitarism, and hormone excess produces the prolactinoma, with amenorrhoea, galactorrhoea and impotence, acromegaly or Cushing disease. A prolactinoma is treated first with *cabergoline or bromocriptine*, not surgery. The endocrine work-up measures prolactin, IGF-1, cortisol and ACTH, thyroid function, gonadotropins and sodium, with formal visual fields.

A **Rathke cleft cyst** is usually incidental and observed. The **craniopharyngioma** has two peaks, in children and in late middle age, is calcified and cystic, and is hard to remove completely. **Meningiomas** arise from the tuberculum, planum and olfactory groove and show a "dural tail". An **empty sella** is associated with idiopathic intracranial hypertension and spontaneous CSF leaks.

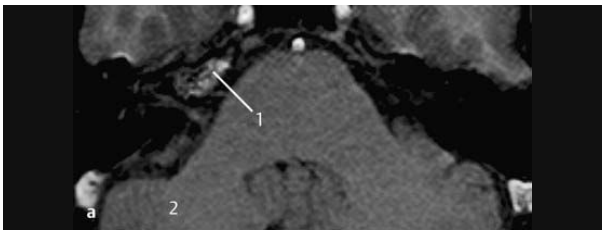
3.3 Clivus and petrous apex

The **chordoma** arises from notochordal remnants at the spheno-occipital synchondrosis in the midline. It is bright on T2 with bone destruction, causes diplopia from a VI palsy and headache, and is treated by maximal resection followed by high-dose, often proton, radiotherapy; recurrence is common. The **chondrosarcoma** arises at the petroclival synchondrosis, off the midline, and has a better prognosis.

At the petrous apex, the cholesterol granuloma is bright on both T1 and T2; cholesteatoma, mucocele and petrous apicitis also occur, the last producing *Gradenigo syndrome* of otorrhoea, VI palsy and retro-orbital pain. Asymmetric pneumatisation or retained fluid can only *look* like a lesion, and MRI settles the question.

IMAGING IN THE APP 1 case

Petrous apex lesions and pseudotumours on imaging.



Pseudotumors in the Apex of the Petrous Bone

Tumours · 12 images

3.4 Lateral skull base and posterior fossa

Here lie the paragangliomas (glomus jugulare, tympanicum and vagale), vestibular schwannoma and other cerebellopontine angle lesions, temporal bone carcinoma, schwannomas of the facial and lower cranial nerves, meningiomas, and the endolymphatic sac tumours of von Hippel–Lindau syndrome.

MRI SIGNAL OF CEREBELLOPONTINE ANGLE AND PETROUS APEX LESIONS

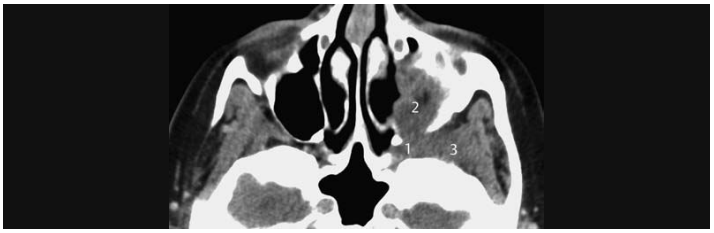
LESION	T1	T1 WITH GADOLINIUM	T2	DIFFUSION
Vestibular schwannoma	Iso or low	Marked enhancement	Moderately high	—
Meningioma	Iso or low	Marked, with a dural tail	Variable	—
Epidermoid (cholesteatoma)	Low	None	High	*Restricted* (bright)
Arachnoid cyst	Low (like CSF)	None	High (like CSF)	Not restricted
Cholesterol granuloma	*High*	None	High	—
Lipoma	High (suppressed by fat saturation)	None	Intermediate	—

3.5 Infratemporal fossa

Primary infratemporal tumours are sarcomas, neurogenic tumours and salivary tumours, but more often the fossa is invaded from its neighbours by nasopharyngeal carcinoma, maxillary sinus cancer, juvenile angiofibroma or parotid cancer. The signs are *trismus* from invasion of the pterygoids, *numbness of V3* over the lower lip and chin, serous otitis media from Eustachian tube invasion, and a mass deep to the ramus.

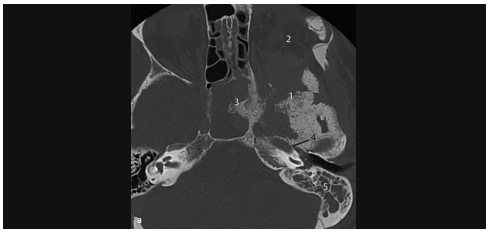
IMAGING IN THE APP 2 cases

Malignancy of the pterygopalatine and infratemporal fossae, and a meningioma destroying the skull base.



Malignancy of the Pterygopalatine and Infratemporal Fossa

Tumours · 2 images



Meningioma with Skull Base Destruction

Tumours · 2 images

4 CHAPTER FOUR Assessment

Nerves, images, tissue.

4.1 Clinical picture

Skull base disease announces itself through whichever structure it reaches first. In the nose it causes unilateral obstruction, bleeding and anosmia; in the eye proptosis, diplopia and visual loss; in the face numbness of V2 or V3 and pain, which suggest *perineural spread*; and in the jaw trismus. In the ear, a unilateral conductive loss from a *middle ear effusion* in an adult means the nasopharynx must be examined and biopsied, and pulsatile tinnitus suggests a vascular tumour. In the throat,

hoarseness, dysphagia and aspiration mean IX to XI are involved, and tongue weakness means XII. CSF rhinorrhoea and headache complete the picture.

The examination is a complete cranial nerve examination, nasal and nasopharyngeal endoscopy, otomicroscopy, and examination of the larynx and neck.

JUGULAR FORAMEN SYNDROMES

- **Vernet syndrome:** IX, X and XI together, from jugular foramen lesions (paragangliomas, schwannomas, metastases).
- **Collet–Sicard syndrome:** adds XII.
- **Villaret syndrome:** adds Horner syndrome, from the retroparotid space.

4.2 Imaging and tissue

CT and MRI are complementary and both are needed. **CT** shows bone detail: erosion, remodelling and widening of foramina. **MRI** with gadolinium and fat suppression shows soft tissue, dura, brain, orbit and perineural spread. Dural invasion is suggested by dural thickening of more than 5 mm or nodular thickening, and periorbital invasion by enhancement or enlargement of the extraocular muscles, stranding of the orbital fat and a nodular interface. **Angiography** is used for vascular tumours and embolisation, and *balloon test occlusion* of the internal carotid when it may have to be sacrificed; **PET-CT** stages disease and detects recurrence.

Most lesions need tissue, often by *endoscopic biopsy*, but not all: paragangliomas, schwannomas, typical meningiomas and juvenile angiofibromas can be diagnosed on imaging alone.

4.3 Unresectability

A tumour is generally unresectable when it invades the brainstem or spinal cord, both internal carotids, both cavernous sinuses or extensive brain parenchyma. Involvement of the optic chiasm, distant metastases and poor general health are relative contraindications. Lymphoma and other chemo- or radiosensitive tumours are not resected at all.

5 CHAPTER FIVE Approaches

Endoscopic and open.

5.1 Endoscopic endonasal surgery

Endoscopic skull base surgery works through both nostrils with the "*binostril, four-handed*" technique, using a wide sphenoidotomy and removal of the posterior septum, with ethmoidectomy, medial maxillectomy, transpterygoid access or a Draf III as needed. The **transsphenoidal** route reaches the sella and is the standard for pituitary surgery; the **transplanum and transtuberculum** routes reach the suprasellar region; the **transcribriform** route exposes the anterior skull base from the frontal sinus to the planum between the orbits; the **transclival** route reaches the clivus down to the foramen magnum and odontoid; and the **transpterygoid** route opens the pterygopalatine and infratemporal fossae, Meckel's cave and the petrous apex.

The approach avoids facial and scalp incisions and brain retraction, gives magnified and angled views and speeds recovery, and for selected sinonasal cancers survival is comparable to open craniofacial resection. Its limits are lateral extension beyond the mid-orbit, involvement of the orbit, the skin or the anterior or lateral wall of the frontal or maxillary sinus, dural or brain invasion lateral to the orbit, and encasement of the carotid.

OPERATIVE GUIDES IN THEATRE 3 operations

Anterior cranial base, transpterygoid and petrosal endoscopic approaches.

Transcranial and Endonasal Approaches to the Anterior Cranial Base
Anterior cranial base
Endoscopic Endonasal Transpterygoid Approaches to the Middle Cranial Fossa
Transpterygoid
Endoscopic Petrosal Approaches

5.2 Reconstruction

Small defects are closed with free grafts of fascia, fat or mucosa in several layers, an *inlay* graft beneath the dura and an *onlay* graft over the bone. Large defects need

vascularised tissue. The *nasoseptal (Hadad–Bassagasteguy) flap*, pedicled on the posterior septal artery from the sphenopalatine artery, has cut postoperative leak rates from over 20% to about 5%, and a "rescue flap" incision at the start of surgery preserves its pedicle in case it is needed. Inferior and middle turbinate flaps, the *pericranial flap* (open or endoscopically harvested), the temporoparietal fascia flap and free flaps for open defects are the alternatives. Lumbar drains are used for selected high-flow leaks, and the patient avoids nose blowing, straining and mask ventilation.

OPERATIVE GUIDES IN THEATRE 2 operations

Reconstruction after skull base surgery, and endoscopic repair of CSF leaks.

Reconstruction After Skull Base Surgery
Skull base reconstruction
Endoscopic Management of Cerebrospinal Fluid Leaks

5.3 Open approaches

Craniofacial resection combines a transfacial route, by lateral rhinotomy or midface degloving, with a bifrontal craniotomy to remove the cribriform plate, ethmoids and involved dura en bloc, and reconstructs with a *pericranial flap* on the supraorbital and supratrochlear vessels. The orbit is exenterated if the periorbital fat, extraocular muscles or orbital apex are invaded.

Fisch's **infratemporal fossa approaches** close the ear canal and remove the middle ear. **Type A** reaches the jugular foramen and infralabyrinthine region with anterior rerouting of the facial nerve, and is used for glomus jugulare; **type B** reaches the clivus and petrous apex; and **type C** the parasellar region, nasopharynx and pterygopalatine fossa.

Other routes include the lateral (preauricular) subtemporal–infratemporal approach, which removes the zygoma and condyle; the far lateral approach to the foramen magnum and lower clivus; the translabyrinthine, retrosigmoid, middle fossa, transotic and transcochlear approaches to the temporal bone and cerebellopontine angle, described with vestibular schwannoma and temporal bone tumours; and the maxillary swing, transpalatal and midface degloving routes to the nasopharynx.

OPERATIVE GUIDES IN THEATRE 3 operations

Approaches to the infratemporal fossa, the maxillary swing and transpalatal routes, and endoscopic nasopharyngectomy.

Surgical Approaches to the Infratemporal Fossa
Infratemporal fossa
Maxillary Swing/Transpalatal/Midface Degloving With Le Fort I Osteotomy Approach for Nasopharyngectomy
Endoscopic Nasopharyngectomy

5.4 Complications

CSF leak is the commonest complication, and meningitis may follow. **Tension pneumocephalus** presents with sudden confusion and a falling conscious level, and CT shows the "*Mount Fuji sign*". **Carotid injury** is managed by packing with pressure, then a crushed muscle patch, then urgent angiography to look for a pseudoaneurysm. Stroke, cranial nerve deficits and visual loss can occur, sellar surgery can cause hypopituitarism and diabetes insipidus (so sodium and urine output are monitored), and the nose is left with crusting, synechiae and anosmia. Trismus and wound or flap failure complete the list.

Overall five-year survival for skull base malignancy is about 50%. Clear margins are the strongest predictor of survival, and postoperative radiotherapy is given for high-risk features.

RECONSTRUCTION OF THE ANTERIOR SKULL BASE (OXFORD TEXTBOOK)

The aim is a durable barrier between the nose and the intradural space to prevent CSF leak, pneumocephalus and meningitis. **Multilayer closure** is standard: an *inlay* graft (usually fascia lata, sometimes with fat) tucked 5 to 10 mm under the dural edges for a watertight seal, then an *onlay* of free mucosa or, preferably, a **vascularised flap**, secured with sealant and supported by dressings or a balloon. Mucosa is stripped from the bone under the onlay so that it adheres and no mucocoele forms.

The choice depends on defect size and site, whether the intraoperative leak is **low-flow** (dura opened without entering a cistern or ventricle) or **high-flow** (a cistern or ventricle opened), and patient factors such as previous radiotherapy, long-term steroids and malnutrition. Small, low-flow leaks close reliably with free grafts, for example the *bath-plug* technique with a fat plug and a free mucosal onlay. Large or high-flow defects, high-risk sites and poor healers need a vascularised flap. The **Hadad–Bassagasteguy nasoseptal flap**, on the posterior septal branch of the

sphenopalatine artery, transformed endoscopic skull base surgery by cutting leak rates; it should be planned (or its pedicle preserved as a "rescue" flap) before the sphenoidotomy. Alternatives on sphenopalatine branches are inferior and middle turbinate flaps and lateral nasal wall flaps; regional options are the **pericranial flap**, on the supraorbital and supratrochlear vessels and passed through a window above the frontal sinus, and the temporoparietal fascia flap.

Lumbar drains reduce pressure on the repair, but with reliable multilayer closure and flaps many surgeons now use them only to treat a postoperative leak, for high-flow defects or in patients with raised intracranial pressure, or to give intrathecal fluorescein.

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Contents of the cavernous sinus;** why VI fails first; the skull base foramina.
- **The pterygopalatine fossa:** its communications and what spreads through it.
- **The parapharyngeal space:** prestyloid versus poststyloid, medial displacement of the tonsil, never biopsy through the mouth, the transcervical approach.
- **Adult unilateral middle ear effusion:** examine the nasopharynx.
- **Chordoma versus chondrosarcoma;** the MRI table for petrous apex and cerebellopontine angle lesions.
- **Endoscopic endonasal approaches,** their limits, and the nasoseptal flap and its pedicle.
- **Craniofacial resection;** the Fisch infratemporal fossa approaches.
- **Complications:** CSF leak, pneumocephalus, carotid injury.

6.2 Common pitfalls

COMMON PITFALLS

- Taking a transoral biopsy of a parapharyngeal mass.
- Treating a prolactinoma surgically before trying a dopamine agonist.
- Missing a VI palsy or facial numbness as signs of skull base spread.
- Calling an asymmetric petrous apex "a lesion" without MRI.

Written for this app from Oxford Textbook of Otolaryngology ch. 44, 45, Oxford Specialist Handbook ch. 19, K. J. Lee ch. 34, 49. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

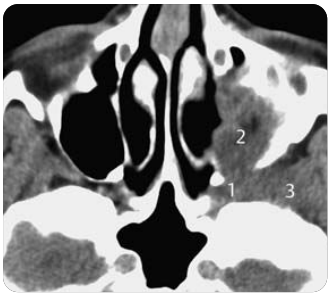
IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

INFRATEMPORAL FOSSA & TMJ · TUMOURS

Malignancy of the Pterygopalatine and Infratemporal Fossa



CT, axial with intravenous contrast enhancement. In the pterygopalatine fossa, a lesion is seen (1) with bony destruction and invasion into the…

TEMPORAL BONE & SKULL BASE · TUMOURS

Pseudotumors in the Apex of the Petrous Bone

IMAGING FINDINGS

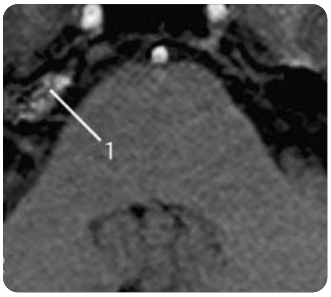
- Asymmetric marrow: bright on T1, suppresses with fat saturation, not expanded
- Trapped fluid: bright on T2, intact septa
- Cholesterol granuloma: bright on T1 and T2, expansile

TOP DIFFERENTIALS

- Cholesteatoma or epidermoid (DWI bright)
- Petrous apicitis (destruction, enhancement)
- Carotid aneurysm

CHECKLIST

- Compare with the other apex
- CT for expansion and cortex



MRI, T1-weighted, axial. T1-weighted image shows that the lesion (1) has high signal intensity compared with brain tissue (2). The contralateral…



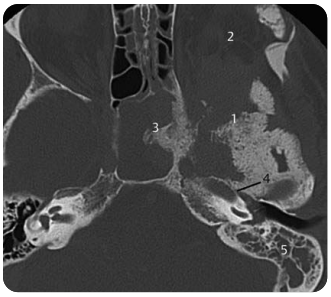
Axial bone CT reveals an expansile cholesterol granuloma in the left PA with marginal bone dehiscence present. The right PA is well…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1152.

PARANASAL SINUSES · TUMOURS

Meningioma with Skull Base Destruction

- On CT, the bone changes associated with hyperostosing meningioma en plaque may be confused with (polyostotic) fibrous dysplasia. However, in meningioma the cortical margins of the affected bones are not well delineated.
- In fibrous dysplasia, the cortical margins are preserved (compare with section on fibrous dysplasia). Cranial nerves and intracranial vasculature may be encroached on by the lesion. Total surgical removal is rarely possible.



CT, axial bone window. On the left side, osseous changes are observed in the area of the anterior fossa (1), orbit (2), and sphenoid sinus (3). The bones…